

The relationship between ABO incompatibility and HLA tissue compatibility during pregnancy losses caused by immunological reasons

Gultakin Aliyeva Budaq

**Nakhchivan State University-docent of medical faculty, Nakhchivan AR
Maternity Center, Physician of Department of Perinatal diagnosis**

Recurrent miscarriage is usually considered after 2 or 3 pregnancy losses (PL). More profound studies should be carried out, if woman is under 30 with 3 Pregnancy Losses or if she is over 30 with 2 Pregnancy Losses.

Fetal loss syndrome is the spontaneous abortion occurring in different periods of pregnancy. This happens for various reasons. For example, structural changes in reproductive system are 88% of the reasons of fetal loss.

Hypothetically, in the proposed experimental group these reasons are shown as following: miscarriage during 5-6th weeks of pregnancy is caused by genetic and immunological reasons, in 7-9th weeks mainly by normal reasons, in 9-10th weeks by autoimmune processes, after 16th week by cervix abnormalities, infectious reasons (TORCH) and thrombocytopenia.

Recurrent pregnancy loss is often considered as a pathological syndrome that caused by polyetiologic symptom complex. Recurrent miscarriage makes 16-20% of all pregnancies.

Purpose of the research is to learn the connection between ABO incompatibility and hidden sensitization in the treatment of fetal loss syndrome with HLA tissue (in 3-4 locus) compatibility.

The objects of research: 2 women in Nakhchivan Maternity Center with fetal loss syndrome. They have compatibility with their husbands in 3-4 locus of HLA tissue.

The location of research: Nakhchivan Maternity Center, Nakhchivan State University

To understand the etiology of recurrent miscarriages we have to pay attention to changes in female organism before pregnancy, during fertilization, nidation of fertilized egg-blastocyst.

Before pregnancy endometrial changes prepare endometrium for implantation of blastocyst. This preparation occurs functionally. At the time between fertilization and nidation of blastocyst endometrium is immunologically prepared.

After nidation in the endometrium starts immune response to the pregnancy. In other words, in case of disorders in any of these mechanisms-both pre-pregnancy preparation, conception and nidation, as well as the subsequent period of nidation, there can be early and late pregnancy losses. Thus, physiological processes are disordered, if functional and organic changes are not normal in endometrium before pregnancy. First of all, it can be a result of acute and chronic inflammatory processes, for example.

Immunological reasons causing miscarriages are divided into the following 2 groups:

1. Autoimmune processes- abnormal response of female organism to her own tissues
2. Alloimmune processes- immune response to foreign tissues.

Miscarriages caused by autoimmune processes reach 10 %.

In 1985 Dr. Graham Huges showed the connection between antiphospholipid antibodies and coagulation in the blood. This problem was investigated in Russia in the same year. In autoimmune disease immune system does not distinguish between healthy tissue and antigens. As a result, the body sets off a reaction that destroys normal tissues.

APS (antiphospholipid syndrome) – is an autoimmune process that manifests arterial and venous thrombosis, fetal loss, neurological and other disorders.

APS includes 6 diseases and recurrent pregnancy loss is ranked 5th in the group. Antiphospholipid antibodies are formed of various immunoglobulins which were raised against (-) charged phospholipids on the cell surface.

Diagnostic tests in APS are based on coagulation process and show the presence of antibodies, are divided into 2 groups:

1. Phospholipid-dependent coagulation tests.
2. The APTT- activated partial thromboplastin time.

These tests are guided by the measurement of prothrombin activator complex. The presence of antiphospholipid antibodies detain this activation. In other words, in all pregnancy losses APS should be taken into account. From the viewpoint of

obstetric the classic triad of APS is recurrent pregnancy loss, thrombocytopenia and thrombosis. For APS diagnosis the tests with an interval of 6 weeks should respond positively 2 times.

The term Alloimmune shows the difference between the same species.

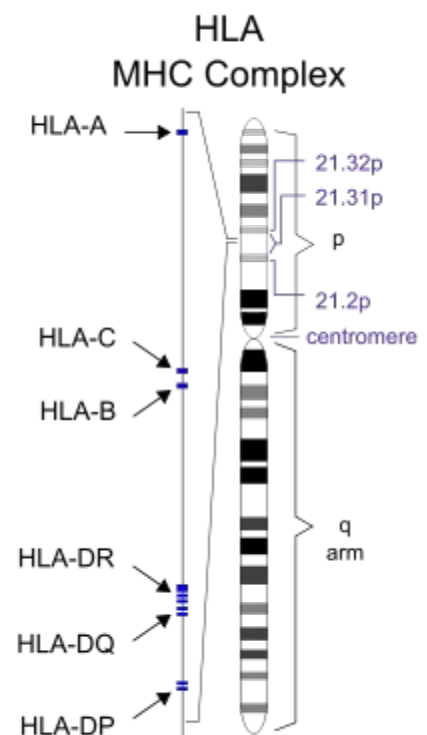
Protein antigens are specific receptors that ensure histological compliance in all cells of the body. The structure of these receptors is unique to each person. Due to this uniqueness immune system can differentiate own and foreign tissues. HLA (Human Leukocyte Antigen) or MHC (Major Histocompatibility Complex) is main complex of tissue and histological compliance and protein based molecule. It is located in all the cells of body, mostly on the surface of leukocytes. HLA controlled by the genes located on chromosome 6. They are divided into 2 groups:

Class I –gene loci A, B, C

Class II – gene locus D (3 sublocus DR,DP,DQ)

HLA plays special role in the acceptance or rejection of tissue and organs in organ and tissue transplantation. If the donor and recipient tissue compatibility is higher, then transfer process will be completed successfully.

The fetus is an alien body for mother. Woman should launch some immunological reaction to adapt this estrangement and complete pregnancy. In other words, unlike organ transplantation the lack of similarity and majority of diversity of HLA protein structures between mother and fetus is a main criteria for normal development and termination of pregnancy. In the first week of pregnancy, the mother's body produces antibodies against the fetal tissue antigen. In the most cases, they arise against Class II antigens. So, if mother and father's Class II antigens are similar, it is regarded as similarity of the fetus to the mother. There is no protective and blocking clones of immune cells. As a result of insufficient stimulation of antibody-antigen pregnancy losses or severe toxemia happens. If 2 antigen with HLA system are similar, then the risk of fetal loss reaches 100%. The factors playing blockade role in female organism in HLA compatibility are not synthesized and divided, growth and development of placental cells stop. In other words, placenta cells die.



It is important to do HLA typing for accuracy of diagnosis. For this, blood cells that fulfill immunological functions-leucocytes separated from blood have been taken from mother and father's ulnar vein. HLA typing is carried out with PCR (Polymerase chain reaction).

However, diagnosis of pregnancy losses caused by alloimmune processes is expensive and conducted in large cities, in clinics equipped with modern technology laboratories. If there are two or more gene loci compatibility of husband and wife, probability of pregnancy loss is 100 %.

Treatment is done with active and passive immunization. Active immunization is carried out with father's or donor's concentrated mass of the lymphocytes, immunized leukocyte solutions. Lymphocyte Immunization Therapy (LIT) is a procedure whereby father's or donor's concentrated mass of lymphocytes is injected into the skin of woman. LIT method increases woman immune response for thousand times. Lymphocyte Immunization Therapy is repeated several times in the 6-8th days of menstrual cycle and throughout the pregnancy. Woman is taught to recognize her husband's cells with this method.

However, among scientists, especially immunologists, still remain dual view about LIT method. Some group of scientists talk about high effect of LIT, while the other group claim for possibility of completing pregnancy with passive immunization.

In passive immunization immunoglobulin is transferred intravenous in the day of ovulation and 3-4 times throughout the pregnancy that stimulates woman's immune response. In HLA compatibility normal immunoglobulin is recommended every other day every month with 50ml x 3. Geotar 2007.

There are results of examination and treatment of two women who have had two pregnancy losses in early weeks.

Patient Q.S., aged 30, 5 years in marriage, has had 3 miscarriages in 6-7th weeks of her pregnancy, uterus cavity has been curettaged.

Spouses blood group belong to the O(I) Rh⁺ . In the Class II HLA typing

HLA-A ⁺	HLA- B ⁺⁺	HLA-C ⁺	HLA - DR ⁺	
A ⁺ - 01	B ⁺ - 51	CW ⁺ - 02	D RBI ⁺ - 13	}
woman				
A ⁺ - 03	B ⁺ - 27	CW ⁺ - 14	D RBI ⁺ - 09	
HLA-A ⁺	HLA- B ⁺⁺	HLA-C ⁺	HLA - DR ⁺	
A ⁺ - 03	B ⁺ - 51	CW ⁺ - 14	D RBI ⁺ - 15	}
husband				
A ⁺ - 11	B ⁺ - 52	CW ⁺ - 12	D RBI ⁺ - 04	

During husband and wife's HLA tissue type examination results comparison we can see 3 antigene type compatibility.

A⁺ - 03 B⁺ - 51 CW⁺ - 14

Passive immunization did not effect and that's why LIT was applied. After this treatment pregnancy ended with the birth of 3100 gr. girl on the date of 19.03.2013.

Another patient R.X., aged 31, 5 years in marriage, has had 4 miscarriages in 7-8th weeks of her pregnancy, uterus cavity has been curretaged.

In HLA tissue type examination there were 4 compatible antigens, 50 % compatibility . Therefore, she was advised to conduct LIT with her husband's lymphocytes in BioMed Medical Center. However, the husband and wife refused the offer and decided to have treatment in Nakhchivan.

DR B₁ DQ A₁ DQ B₁

R.X. “04 ”11 “0301- “0501 0305 0301/4 woman

R.R. “04 ”11 “0301-“0501 0302 0301/4
husband

In addition to the similarity of detected Alloimmunologic marker 04“11 ; “0301 ; “0501 ; “0301/4, CMV , herpes, toxoplasmosis, Chlamydia and ABO-system-direct Coombs test on the record of 1:256 was found, which led to the woman and husband’s secret sensitization. ABO incompatibility is becoming dominant system during Rh incompatibility and it gives lighter clinical picture to the hemolytic disease of fetus. Therefore, we decided to start the examination and treatment thinking about the positive result from the relationship between ABO incompatibility and HLA tissue compatibility of woman and husband’s 3-4 loci. Moreover, we have not forgotten that some microorganisms trigger autoimmune, isoimmune, alloimmune processes during pregnancy, some pathogens mimicry blood groups. We completed the course of treatment. Husband and wife who refused LIT therapy passed health program before pregnancy. They received desensitization therapy for ABO incompatibility. Thus, the factors affecting early pregnancy loss, some of the reasons of miscarriages after nidation- endometrium efficiency and non-acceptance and rejection of blastocyst by mother’s organism have been eliminated. The husband and wife received specific therapy for viral and bacterial infections, ABO incompatibility treatment and they have 2 daughters at the moment. Sometimes it is considered in obstetric practice that bacterial and viral infections don’t cause recurrent pregnancy losses, patients are advised not to exceed the examination. However, it should be noted that endometrium damaged by viral and bacterial infections causes negligence of nidation, deterioration of nutrition of embryo. Thus, acute and chronic inflammatory process directly influences pregnancy loss with negligence of implantation and deterioration of nutrition, indirectly with triggering allo and isoimmune processes.

Thus, during the assessment of pregnancy loss caused by immunological reasons it became clear that for the first woman having compatibility with ABO system and HLA tissue in 2-3 loci more serious LIT treatment was required. HLA, ABO, Rh and other rare blood compatibility and incompatibility, as well as secret sensitization, viral, bacterial infections should not be overlooked. Inflammatory process causes specific or non-specific inflammation – endometritis and it

disorders the physiological preparation for pregnancy. The pregnancy has been completed after creation of strong immune response with LIT therapy.

On the other hand, HLA tissue compatibility in 4 loci and ABO incompatibility (ABO titer 1/256) were revealed in the second woman. We treated dominant ABO incompatibility and achieved good immune status.

Conclusion:

1. If wife and husband's Class II antigens are similar, it is regarded as similarity of the fetus to the mother. There is no protective and blocking clones of immune cells. As a result of insufficient stimulation of antibody-antigen pregnancy losses happen.
2. In case of wife and husband's HLA tissue compatibility in 3-4 loci, ABO incompatibility (ABO titer 1/128, 1/256), it should be noted that bacterial and viral infections mimicry blood groups. The ABO incompatibility having dominant character will be treated easier.
3. Restoration of immune status and completion of pregnancy is achieved by means of the treatment of dominant ABO incompatibility with desensitization therapy.

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